

GENETIC DISORDERS

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Aneuploidies

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Abnormal number of chromosomes.

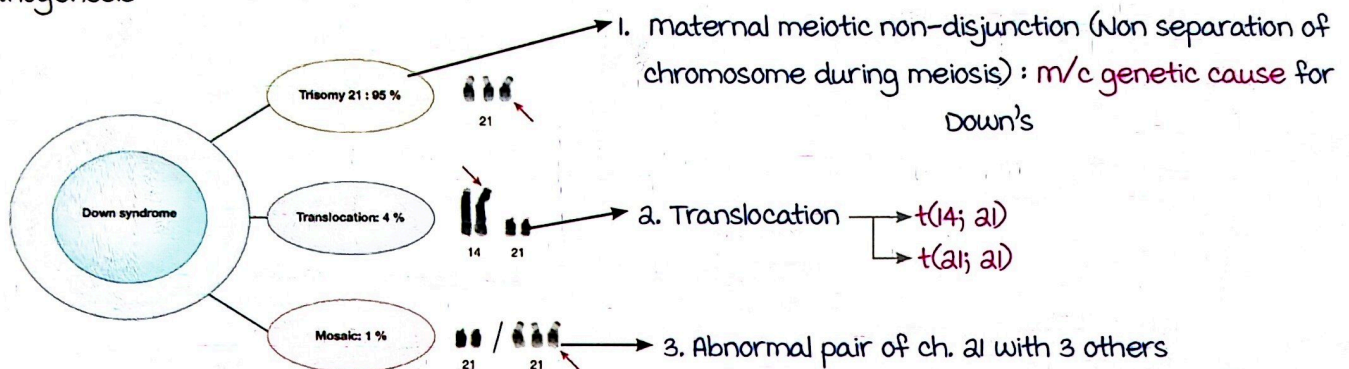
TRISOMIES :

Presence of **three chromosomes** in place of a pair.

Down Syndrome (Trisomy 21) :

- m/c trisomy.

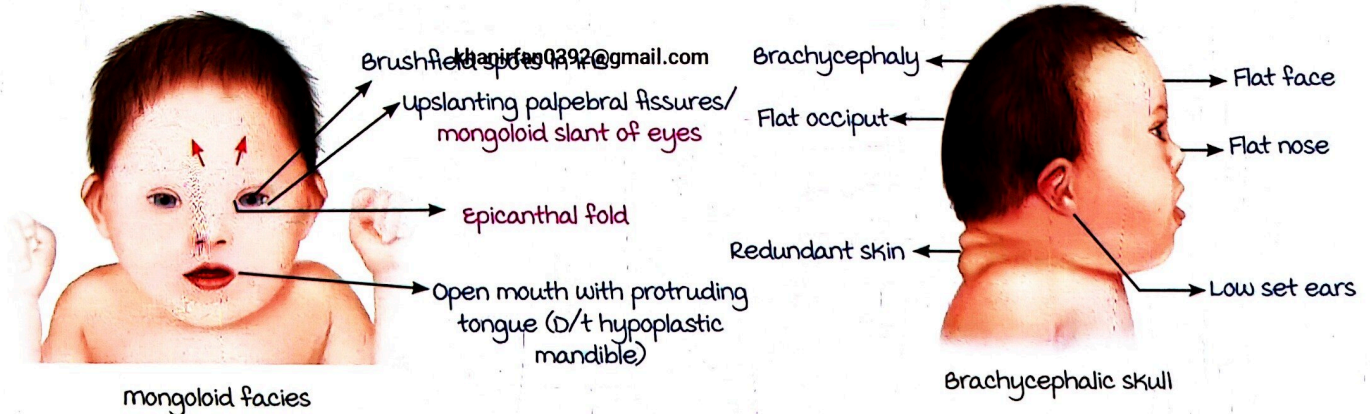
Pathogenesis :



Chromosomal abnormalities found in down's syndrome

Clinical Features :

1. Facial abnormalities/face and skull.



2. Limb abnormalities :

- Palms : **Simian crease** (Single transverse palmar crease).
- 5th/little finger : **Clinodactyly** (Curved finger d/t hypoplasia of phalynx).
- Foot : **Sandal gap** (↑ed gap between 1st and 2nd toe).

Internal manifestations :

- a. **Heart** : Congenital Heart Defect (m/c : 40%).

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- Endocardial Cushion/atrioventricular septal defect.
- m/c cause of morbidity/mortality.

b. Brain : Low IQ, Hypotonia.

c. GI defects :

- Intestinal atresia (m/c : Duodenum) → Intestinal obstruction.
- Others : Hirschsprung disease/aganglionosis, annular pancreas.

Other associations :

1. Hypothyroidism : Congenital or acquired.

- Autoantibodies found in older children (Eg: Antithyroid peroxidase antibodies)

2. Leukemia :

- Acute leukemia → ALL: m/c overall
→ AML - m7 (megakaryoblastic) : m/c in < 3 yrs of age

3. Presenile alzheimer's disease : 35-40 years of age.

Trisomy 21 → Overexpression of amyloid precursor protein (APP) gene → Amyloid ↑ → Deposited in the brain → Presenile Alzheimer's.

4. Atlanto-axial subluxation/instability (10-30%)

↓ upon hyperflexion (Eg: During sports)

Sudden compression of spinal cord.

- Diagnose : Lateral X-ray of neck.
- Advice : Avoid contact sports.

Recurrence risk :

- General : 1%.
- Maternal age > 35 years : upto 4%.
- Parents with robertsonian translocation.

Translocation	In mother	In Father
t(14; 21)	10%	5%
t(21; 21)	100%	100%

Edwards Syndrome : Trisomy 18 :

- and m/c trisomy.

External manifestations :

microphthalmos
(Short palpebral fissures)

Short neck

Limited hip abduction

Rocker bottom feet

Occiput or back part of skull is prominent

Dysplastic or malformed ears

Clenched hands with overlapping fingers

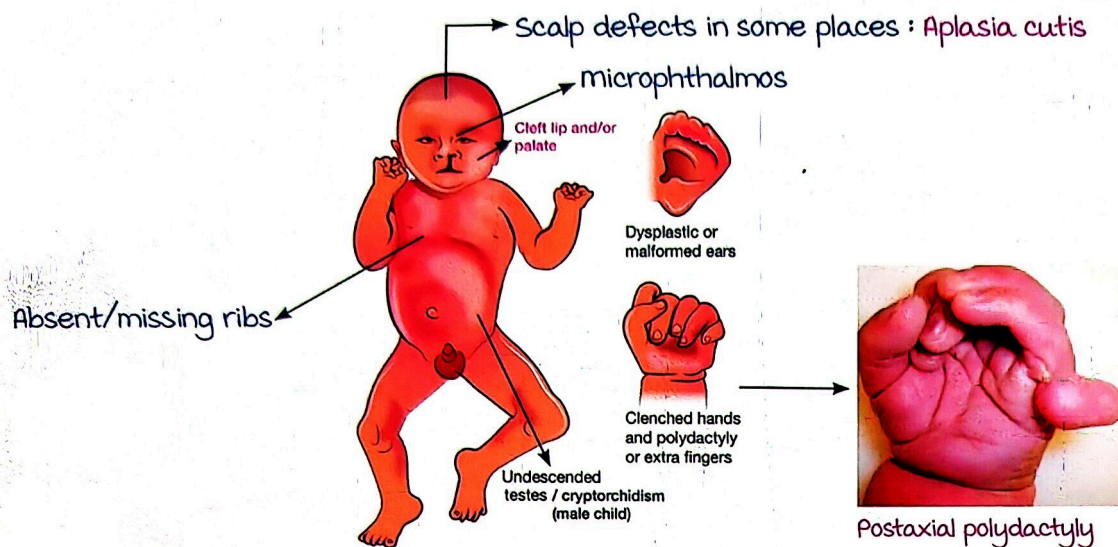
Dorsiflexed toe prominent heels

Internal manifestations :

- CHD (> 90%) : VSD > PDA > ASD.
- Renal: horseshoe kidney.

Patau Syndrome : Trisomy 13 :

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Heart defects (80-90%) : **VSD** > **PDA** > **ASD**.

CNS : **Holoprosencephaly**.

TURNER SYNDROME :

Karyotype : **45, XO/monosomy X** (57%) > mosaicism (29%) > Partial monosomy X (14%)
(monosomy X in some cells) (Ring/iso chromosome X)

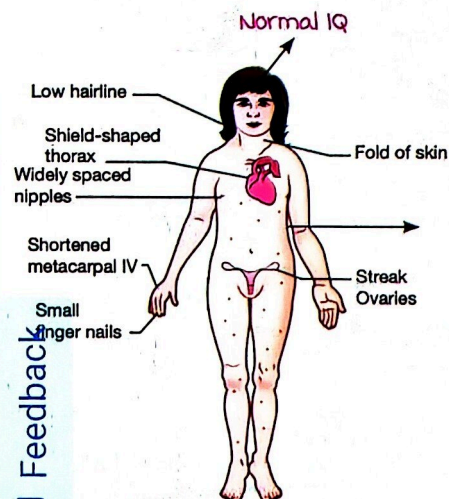
Gender : Female.

Clinical Aspects :

External manifestations :

- Short stature (Adults : < 145 cm).
- Neck → Web neck
→ Loose skin folds

Lymphedema : — may involve foot/hand/neck **cystic hygroma**
— seen during **infancy**



Cubitus valgus : **Carrying angle** > **135°** (Increased)
(Lateral deviation of limbs at the level of the elbow)

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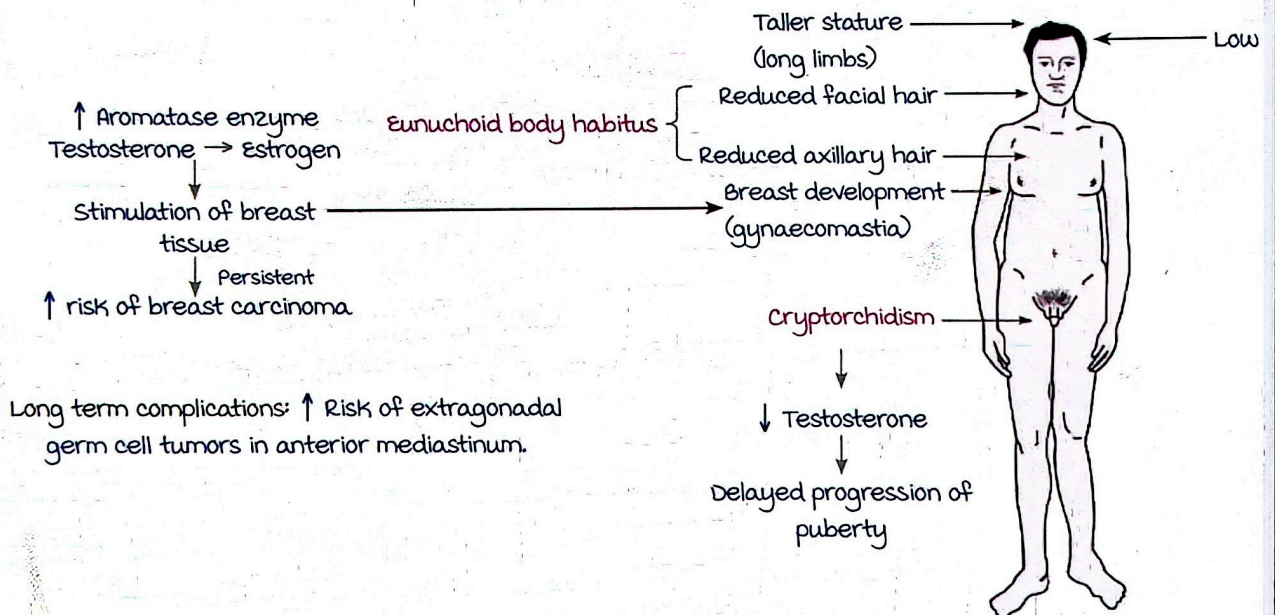
Internal manifestations :

- Heart : Bicuspid aortic valve/coarctation of aorta.
- Kidney (50%) : Horse shoe kidney.
- Delayed puberty & Infertility in adulthood.

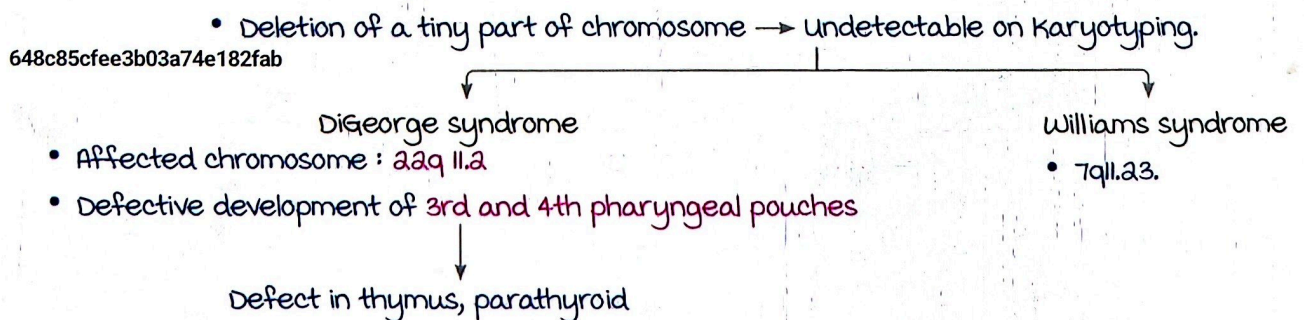
Pseudo Turner Syndrome/Noonan Syndrome :

- Inheritance : Autosomal dominant (males and Females can be affected).
- Gene defect : **PTPN** (Protein tyrosine phosphatase, non-Receptor) gene located on chromosome 12q24.

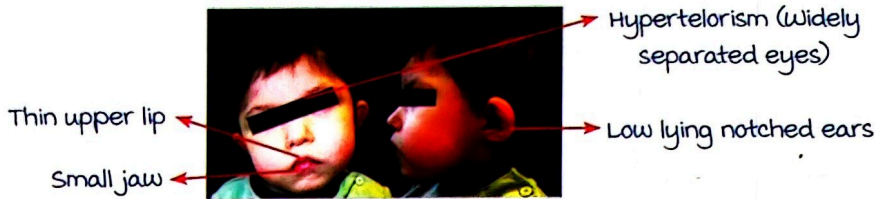
	Turner	Noonan
Gender	Females	Both gender
IQ	Normal	Low
Gonadal defect	Streak ovaries	Cryptorchidism (male)
m/c heart defect	Bicuspid aortic valve, Coarctation of aorta	Pulmonary stenosis

KLINEFELTER SYNDROME :**Microdeletion Syndromes**

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• Clinical features :



• Associations :

C : Cardiac conotruncal defects (Tetralogy of Fallot : m/c)

A : Abnormal face.

T : T cell deficiency (Due to absence of thymus.)

C : Cleft lip/palate.

H : Hypocalcemia (Due to PTH deficiency).

aa : Chromosome aa.

• Clinical features :

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Elfin face

Associations :

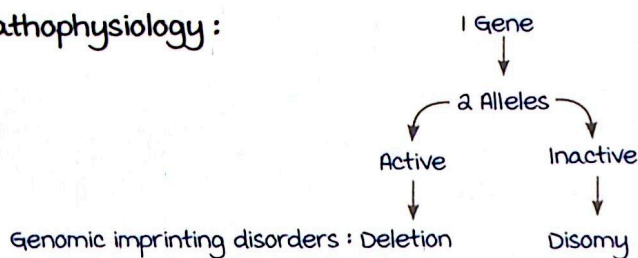
- Hypercalcemia.
- Heart defect : Supravalvular aortic stenosis (m/c).
- Low IQ, learning disabilities.

Genomic Imprinting Disorders

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- Imprinting : Silencing.

Pathophysiology :



Prader-Willi syndrome :

- Genetic defect : Chromosome 15q 11 - 13
 - Deletion of paternal allele
 - Disomy of maternal allele

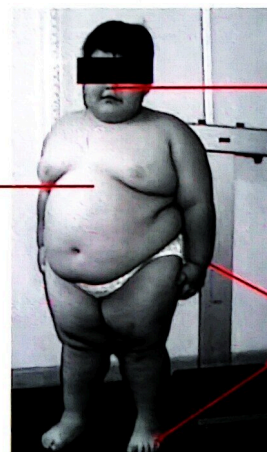
Clinical Features :

1. massive Obesity :

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- ↑ Ghrelin → ↑ appetite → ↑ food intake

2. Short stature : Compared to children of similar age



4. Small eyes :
Almond shaped eyes

3. Small hands
and feet

5. Hypogonadism → Delayed puberty → Low IQ

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Genetic Disorders

Angelman syndrome :

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- AKA happy puppet syndrome.
- Gene defect : 15 q 11 - 13
 - Deletion of maternal allele
 - Disomy of paternal allele

Clinical Features :



- Neurological features :
 - ↑ incidence of seizures.
 - Hypotonia.
 - Low IQ.
 - Speech defects (Communicate in sign language).
 - Abnormal movements of hands & feet : Ataxia

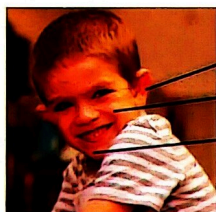
Trinucleotide Repeat Disorders (TRD)

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FRAGILE X SYNDROME :

- CGG repeats
 - < 55 : Normal
 - > 200 : TRD
- Gene defect : FMR1 (Fragile messenger ribonucleoprotein) protein gene.

Clinical Features :



Large ears

Long face

Prominent chin

- macroorchidism (Large testes) : Post puberty.
- ↑ Behavioural abnormalities :
 - ADHD (Attention deficit hyperactive disorder).
 - Autism.

Other associations :

- Connective tissue defects (CTD)
 - Hyperextensive joints
 - High arched palate
 - mitral valve prolapse (mvp)
- Low IQ.